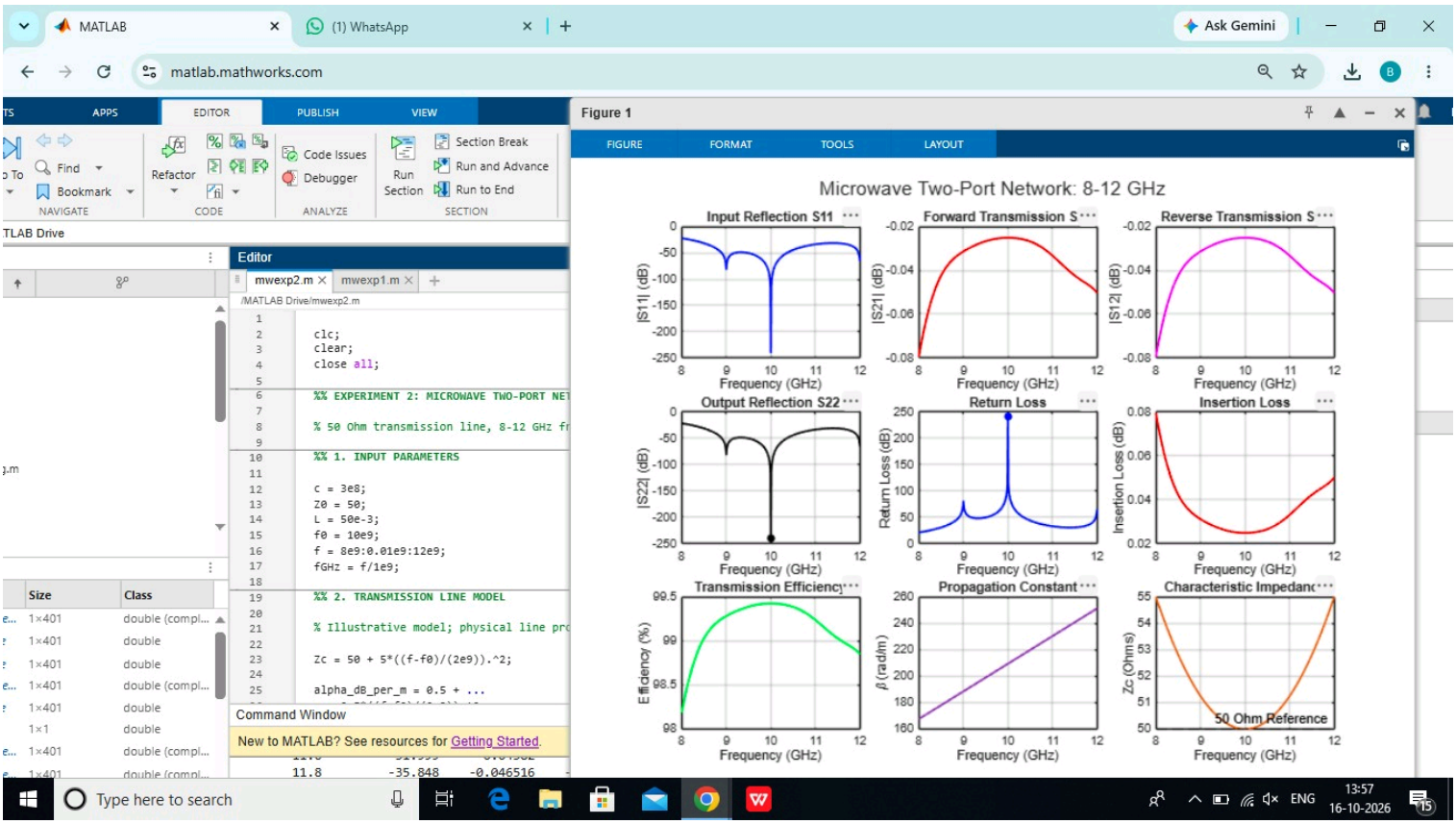

MATLAB
(1) WhatsApp
Ask Gemini
matlab.mathworks.com
APPS
EDITOR
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Refactor
CODE
Code Issues
Debugger
ANALYZE
Run Section
Section Break
Run and Advance
Run to End
SECTION
TLAB Drive
Editor
mwexp2.m
mwexp1.m
/MATLAB Drive/mwexp2.m
clc;
clear;
close all;
%% EXPERIMENT 2: MICROWAVE TWO-PORT NE
% 50 Ohm transmission line, 8-12 GHz f
%% 1. INPUT PARAMETERS
c = 3e8;
Z0 = 50;
L = 50e-3;
f0 = 10e9;
f = 8e9:0.01e9:12e9;
fGHz = f/1e9;
%% 2. TRANSMISSION LINE MODEL
% Illustrative model; physical line pr
Zc = 50 + 5*((f-f0)/(2e9)).^2;
alpha_dB_per_m = 0.5 + ...
Command Window
New to MATLAB? See resources for Getting Started.
Size
Class
1×401 double (compl...
1×401 double
1×401 double
1×401 double (compl...
1×401 double
1×1 double
1×401 double (compl...
11.8 -35.848 -0.046516
Figure 1
FIGURE
FORMAT
TOOLS
LAYOUT
Microwave Two-Port Network: 8-12 GHz
Input Reflection S11
Forward Transmission S
Reverse Transmission S
Output Reflection S22
Return Loss
Insertion Loss
Transmission Efficiency
Propagation Constant
Characteristic Impedanc
|S11| (dB)
|S21| (dB)
|S12| (dB)
|S22| (dB)
Return Loss (dB)
Insertion Loss (dB)
Efficiency (%)
β (rad/m)
Zc (Ohms)
50 Ohm Reference
Frequency (GHz)
Type here to search
ENG
13:57
16-10-2026